Experiment 03 – Lab 2

Types of Prompts and Prompt Engineering Basics

Aim

To study different types of prompts used in ChatGPT, understand prompt engineering, and compare the results obtained from different prompts.

Introduction

A **prompt** is an instruction or question given to ChatGPT to generate a response. Prompt engineering is the process of designing clear and effective prompts to obtain accurate, relevant, and useful results. This experiment covers zero-shot, one-shot, few-shot, instructional, and interrogative prompts.

1. Zero-Shot Prompting

Definition: Asking ChatGPT to perform a task without providing examples.

1. Definition

Prompt: Explain what Artificial Intelligence is.

Sample Output: Artificial Intelligence (AI) enables machines to perform tasks that typically require human intelligence, such as learning, reasoning, and problem-solving.

2. Programming

Prompt: Write a Python program to add two numbers.

```
Sample Output: a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
print("Sum =", a + b)
```

3. Education

Prompt: Explain the water cycle.

Sample Output: The water cycle is the continuous movement of water between Earth's surface and the atmosphere through evaporation, condensation, and precipitation.

4. Summarization

Prompt: Summarize the importance of trees in two sentences.

Sample Output: Trees absorb carbon dioxide, release oxygen, provide habitats for wildlife, help prevent soil erosion, and maintain ecological balance.

5. Creativity

Prompt: Write a short slogan about saving water.

Sample Output: Save Water, Save Life!

Observation: Zero-shot prompting is simple and useful for general questions and common tasks. A specific format should be requested if needed.

2. One-Shot Prompting

Definition: Providing one example to show ChatGPT the desired pattern or format.

1. Sentiment Classification

Prompt: Example: "This movie is wonderful." → Positive
Classify: "The food was delicious."

Sample Output: Positive

2. Text Classification

Prompt: Example: apple → Fruit
Classify: Carrot → ?

Sample Output: Carrot → Vegetable

3. Sentence Transformation

Prompt: Example: "He is happy." → "He is very happy."
Transform: "She is tired."

Sample Output: She is very tired.

4. Email Writing

Prompt: Example: "Send me the notes." → "Could you please send me the notes?"
Rewrite: "Give me the report."

Sample Output: Could you please provide me with the report?

5. Mathematical Pattern

Prompt: Example: 2 → 4 (multiply by 2)
Apply the same rule: 5 → ?

Sample Output: 5 → 10

Observation: One-shot prompting helps communicate the expected response style through one example. It is useful for formatting, classification, and simple patterns.

3. Few-Shot Prompting

Definition: Providing two or more examples before asking ChatGPT to perform a task.

1. Sentiment Classification

Prompt: Examples: "I love this product." → Positive; "The service was terrible." → Negative; "The experience was excellent." → Positive.
Classify: "I hate waiting."

Sample Output: Negative

2. Fruit Classification

Prompt: Examples: Apple → Fruit; Potato → Vegetable; Mango → Fruit; Spinach → Vegetable.
Classify: Banana.

Sample Output: Banana → Fruit

3. Temperature Classification

Prompt: Examples: 5°C → Cold; 20°C → Moderate; 35°C → Hot.
Classify: 10°C.

Sample Output: 10°C → Cold

4. Formal Email Writing

Prompt: Examples: "Give me the file." → "Could you please share the file?"; "Tell me the answer." → "Could you please provide the answer?"
Convert: "Send me the details."

Sample Output: Could you please provide me with the details?

5. Number Classification

Prompt: Examples: 4 → Even; 7 → Odd; 10 → Even; 13 → Odd.
Classify: 18.

Sample Output: 18 → Even

Observation: Few-shot prompting is useful for consistent patterns and formats. Results depend on the quality and relevance of examples.

4. Instructional Prompting

Definition: Giving clear directions about the task, format, length, style, or other requirements.

1. Explanation

Prompt: Explain cloud computing in simple language using three bullet points.

Sample Output: Cloud computing provides computing services over the internet.
It allows users to store files online.
Examples include Google Drive and Microsoft OneDrive.

2. Programming

Prompt: Write a Python program to check whether a number is even or odd. Include comments and explain the logic in two sentences.

```
Sample Output: num = int(input("Enter a number: "))
if num % 2 == 0:
    print("Even")
else:
    print("Odd")

The modulus operator finds the remainder after division by two. A zero remainder means even;
otherwise, the number is odd.
```

3. Essay Writing

Prompt: Write approximately 100 words about the importance of education in simple English.

Sample Output: Education provides knowledge, develops skills, and helps people make better decisions. It can improve employment opportunities and encourage creativity and critical thinking. Education also teaches values such as discipline, respect, and responsibility. An educated society can contribute to economic growth and social development. Therefore, education is essential for building a better future.

4. Table Creation

Prompt: Compare RAM and ROM in a table with three differences.

Sample Output: RAM: temporary/volatile memory; used for currently running programs and data.
ROM: generally non-volatile memory; commonly stores firmware and startup instructions.

5. Study Planning

Prompt: Create a seven-day beginner Python study plan as a table with topics and daily practice tasks.

Sample Output: Day 1: Variables — print a message.
Day 2: Data types — create variables.
Day 3: Operators — perform calculations.
Day 4: Conditions — check even or odd.
Day 5: Loops — print sequences.
Day 6: Lists — store and display values.
Day 7: Revision — build a small calculator.

Observation: Instructional prompts are useful when a specific structure, length, tone, or level of detail is required.

5. Interrogative Prompting

Definition: Asking ChatGPT a question to obtain information, explanations, or solutions. Questions often begin with what, why, how, when, or which.

1. General Knowledge

Prompt: What is Machine Learning?

Sample Output: Machine Learning is a branch of AI that enables computers to learn patterns from data and make predictions or decisions.

2. Science

Prompt: Why is the sky blue?

Sample Output: Air molecules scatter shorter wavelengths of sunlight, such as blue light, more strongly than longer wavelengths.

3. Programming

Prompt: How does a for loop work in Python?

Sample Output: A for loop repeats a block of code for each item in a sequence or iterable.

```
for i in range(1, 6):
    print(i)
```

This prints the numbers 1 to 5.

4. Career Guidance

Prompt: Which programming language should a beginner learn first?

Sample Output: Python is often a good choice because it has readable syntax and supports automation, data analysis, and AI.

5. Environmental Awareness

Prompt: How can students reduce plastic pollution?

Sample Output: Use reusable bottles and bags, avoid single-use plastics, separate waste, recycle, and participate in cleanup activities.

Observation: Interrogative prompts are useful for direct questions and learning concepts. Clear questions generally lead to more relevant answers.

6. Comparison of Prompt Types

Type	Examples	Purpose	Advantage
Zero-Shot	No examples	Perform a task directly	Simple and quick
One-Shot	One example	Show the expected pattern	Establishes a format
Few-Shot	Two or more examples	Demonstrate a pattern	Can improve consistency
Instructional	Examples not required	Specify task requirements	Controls format and detail
Interrogative	Examples not required	Ask a question	Useful for information and explanations

Overall Comparison of Results

Zero-shot prompting suits straightforward tasks; one-shot prompting demonstrates a pattern with one example; few-shot prompting supplies multiple examples for a more consistent pattern; instructional prompting gives control over structure and detail; and interrogative prompting is useful for questions and explanations. For example, a general zero-shot request to explain Python may produce a broad explanation, while an instructional prompt can ask for simple English and a code example.

Note: The outputs in this document are illustrative examples of possible ChatGPT responses, not results from a controlled experiment. Actual responses may vary.

Conclusion

This experiment demonstrates five types of prompts: zero-shot, one-shot, few-shot, instructional, and interrogative. Each serves a different purpose. Understanding these techniques helps users design clearer prompts and obtain more useful AI-generated responses.

Result: Successfully studied five types of prompts, prepared five examples for each type, and compared their purposes and advantages.